Observing Crossing Over

Biology 2e chapter 11 reading connection | Student worksheet / preparation

Name: _____ Class: _____ Date: _____

Before predicting

An ascus contains an ordered set of spores. In a suitable two-color Sordaria cross, classify intact eight-spore asci by their color sequence. Keep the original order while counting. Use only complete, interpretable asci.

Investigation question

What can ordered spore patterns reveal about crossing over?

Prediction and reasoning:

Controls, independent variable, measurement and sampling plan:

How to use the virtual lab

Classify every numbered ascus once. Dark and tan identify the two spore-color types. A reversed color order belongs to the same pattern category. Review your classifications before collecting the field.

Prepare the microscope using its setup controls. Inspect and classify all complete ordered asci. Review the classifications; retain and explain any uncertain observations. Review observations, then click Collect data and continue. Complete the assigned comparison before exploring.

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Label every sample virtual or classroom. Keep independent replicates separate; retain unexpected results. Units: asci

Condition / sample	Repeat	Source	4:4	2:4:2	2:2:2:2
Field A	1				
Field B	1				

Equipment and observations

Record calibration, initial values, specimen selection, ambiguous observations or method changes. Use additional paper for larger classroom counts.

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Sample / field: _____ Source: _____

Item / band / cell	Measurement or classification	Reasoning / uncertainty

For large counts, use tallies with a clear starting position so each specimen is counted once. For microscopy and gels, record calibration and measurement units.

Observing Crossing Over

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Calculate second-division segregation frequency and the approximate map distance.

Why is the percentage divided by two?

How could excluded asci and multiple crossovers affect the estimate?

Explore after the comparison

Propose one additional test. Change one condition or sample choice, collect exploration separately, and explain what the new evidence adds.

Important limits: Original ordered-spore schematics. The simple mapping estimate assumes the scoring model and can miss multiple crossovers; this is gene-to-centromere mapping, not ordering several genes.